Trường Đại học Công nghệ Thông tin
Khoa Mạng máy tính & Truyền thông

Lab
5

BÁO CÁO BÀI THỰC HÀNH SỐ 5
KUBERNETES (K3S)

Môn học: Hệ tính toán phân bố
Lớp: NT533.P11.1

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Phần bên dưới của báo cáo này là bài làm chi tiết của sinh viên thực hiện.

MỤC LỤC

BÀI LÀM CHI TIẾT

1. Triển khai K3s cluster trên 2 thiết bị Raspberry Pi sử dụng K3sup

Vì không có thiết bị Raspberry Pi nên nhóm sẽ triển khai trên máy ảo Ubuntu:

1.1. Trên 2 máy tính trong K3s Cluster

1.1.1. Cài đặt SSH

1.1.2. Cấu hình cho phép SSH với tài khoản root

```
vi /etc/ssh/sshd_config
# Thêm vào file dòng dưới và lưu lại
PermitRootLogin yes
```

1.1.3. Chỉnh sửa hostname cho mỗi máy trong Cluster

- Mỗi máy trong Cluster phải có hostname khác nhau:

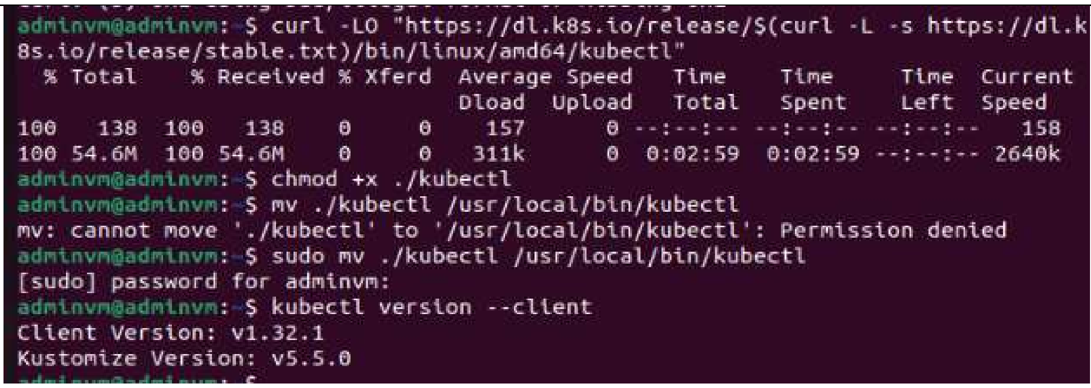
Server	$ hostnamectl set-hostname server $ reboot
Agent	$ hostnamectl set-hostname agent $ reboot

1.2. Trên máy tính Ubuntu

1.2.1. Download, cài đặt và kiểm tra phiên bản Kubectl

```
apt install -y curl
curl -LO "https://dl.k8s.io/release/$(curl -L -s
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
chmod +x ./kubectl
```

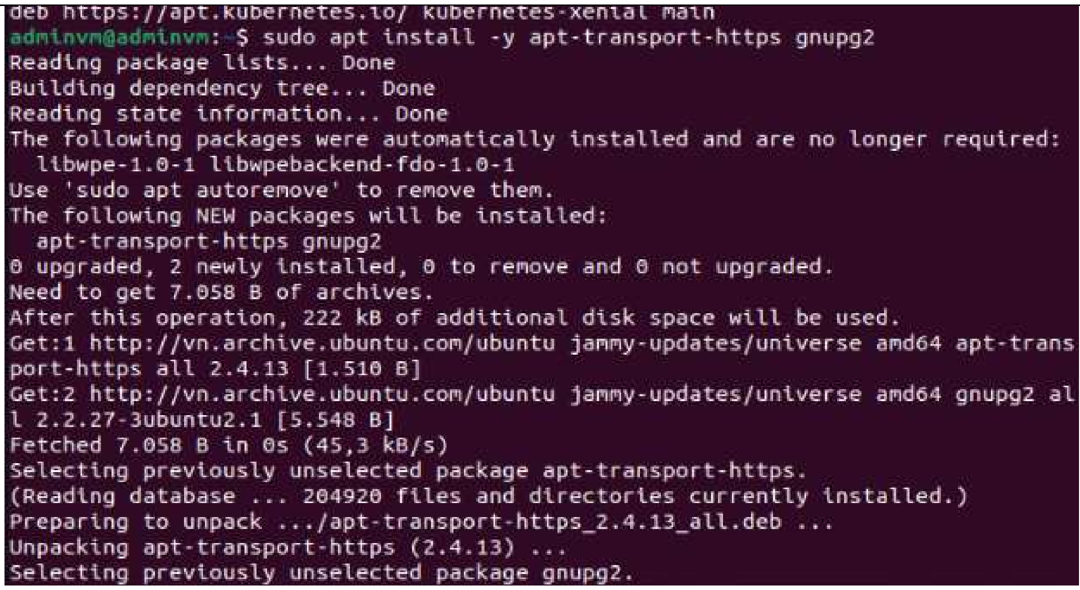
```
mv ./kubectl /usr/local/bin/kubectl
kubectl version –client
```



```
adminvm@adminvm:~$ curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k
8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   138  100   138    0     0    157      0 --:--:-- --:--:-- --:--:--   158
100 54.6M  100 54.6M    0     0   311k      0  0:02:59  0:02:59 --:--:-- 2640k
adminvm@adminvm:~$ chmod +x ./kubectl
adminvm@adminvm:~$ mv ./kubectl /usr/local/bin/kubectl
mv: cannot move './kubectl' to '/usr/local/bin/kubectl': Permission denied
adminvm@adminvm:~$ sudo mv ./kubectl /usr/local/bin/kubectl
[sudo] password for adminvm:
adminvm@adminvm:~$ kubectl version --client
Client Version: v1.32.1
Kustomize Version: v5.5.0
```

1.2.2. Cài đặt các package management

```
apt install -y apt-transport-https gnupg2
curl -s https://packages.cloud.google.com/apt/doc/apt-key.gpg | apt-key add -
echo "deb https://apt.kubernetes.io/ kubernetes-xenial main" | tee -a
/etc/apt/sources.list.d/kubernetes.list
apt install -y kubectl
snap install kubectl --classic
kubectl version –client
```



```
deb https://apt.kubernetes.io/ kubernetes-xenial main
adminvm@adminvm:~$ sudo apt install -y apt-transport-https gnupg2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  apt-transport-https gnupg2
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 7.058 B of archives.
After this operation, 222 kB of additional disk space will be used.
Get:1 http://vn.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 apt-trans
port-https all 2.4.13 [1.510 B]
Get:2 http://vn.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 gnupg2 al
l 2.2.27-3ubuntu2.1 [5.548 B]
Fetched 7.058 B in 0s (45,3 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 204920 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.4.13_all.deb ...
Unpacking apt-transport-https (2.4.13) ...
Selecting previously unselected package gnupg2.
```

1.2.3. Cài đặt K3sup và Arkade:

```
# Cài đặt K3sup
curl -sLS https://get.k3sup.dev | sh
install k3sup /usr/local/bin/
k3sup --help
# Cài đặt arkade
curl -sLS https://get.arkade.dev | sh
arkade –help
```

```
adminvm@adminvm:~$ curl -sLS https://get.k3sup.dev | sh
x86_64
Downloading package https://github.com/alexellis/k3sup/releases/download/0.13.8/
k3sup as /home/adminvm/k3sup
Download complete.

============================================================
  The script was run as a user who is unable to write
  to /usr/local/bin. To complete the installation the
  following commands may need to be run manually.
============================================================

  sudo cp k3sup /usr/local/bin/k3sup


================================================================
  Support Alex's work on K3sup via GitHub Sponsors

  https://github.com/sponsors/alexellis
================================================================
```

```
adminvm@adminvm:~$ curl -sLS https://get.arkade.dev | sh
x86_64
Downloading package https://github.com/alexellis/arkade/releases/download/0.11.3
2/arkade as /home/adminvm/arkade
Download complete.

============================================================
  The script was run as a user who is unable to write
  to /usr/local/bin. To complete the installation the
  following commands may need to be run manually.
============================================================

  sudo cp arkade /usr/local/bin/arkade
  sudo ln -sf /usr/local/bin/arkade /usr/local/bin/ark
```

1.2.4. Kiểm tra SSH Key

ls -l ~/.ssh/id_rsa.pub
#Nếu chưa có key, tạo key mới
ssh-keygen

```
adminvm@adminvm:~$ sudo ls -l ~/.ssh/id_rsa.pub
[sudo] password for adminvm:
ls: cannot access '/home/adminvm/.ssh/id_rsa.pub': No such file or directory
adminvm@adminvm:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/adminvm/.ssh/id_rsa):
Created directory '/home/adminvm/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/adminvm/.ssh/id_rsa
Your public key has been saved in /home/adminvm/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:LoPirSV0dLdEEjDYe0mZpXSYc/NzoMWx9641zOKFyaM adminvm@adminvm
The key's randomart image is:
+---[RSA 3072]----+
|    oo.+B+...    |
|  . .oB=o +.     |
|     .oo+o=...   |
|    ...oo..o...  |
|   . ..  S  o  . |
| . . . .    . *  |
|  o o o .   * B  |
| . =   o   o * . |
|  o..      E o   |
+----[SHA256]-----+
adminvm@adminvm:~$ sudo ls -l ~/.ssh/id_rsa.pub
-rw-r--r-- 1 adminvm adminvm 569 Thg 1  25 01:46 /home/adminvm/.ssh/id_rsa.pub
```

1.2.5. Copy SSH Key vào các máy trong Cluster

```
ssh-copy-id adminvm@192.168.10.131
ssh-copy-id adminvm@192.168.10.134
```

```
adminvm@adminvm:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub adminvm@192.168.10.131
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/adminvm/.ss
h/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
adminvm@192.168.10.131's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'adminvm@192.168.10.131'"
and check to make sure that only the key(s) you wanted were added.

adminvm@adminvm:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub adminvm@192.168.10.134
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/adminvm/.ss
h/id_rsa.pub"
The authenticity of host '192.168.10.134 (192.168.10.134)' can't be established.
ED25519 key fingerprint is SHA256:hSuGu7+ES9a1+8RI2bP0+COkgGFyOXaNg9sVSmw/yM8.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])?
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
The authenticity of host '192.168.10.134 (192.168.10.134)' can't be established.
ED25519 key fingerprint is SHA256:hSuGu7+ES9a1+8RI2bP0+COkgGFyOXaNg9sVSmw/yM8.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
adminvm@192.168.10.134's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'adminvm@192.168.10.134'"
and check to make sure that only the key(s) you wanted were added.
```

1.2.6. Cài đặt K3s lên master

```
export SERVER=192.168.10.131
k3sup install --ip $SERVER --user adminvm --ssh
export KUBECONFIG=/home/ubuntu/kubeconfig
# Thêm dòng trên vào cuối file .bashrc
nano ~/.bashrc
```

```
  GNU nano 6.2                              /root/.bashrc *
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

export KUBECONFIG=/home/ubuntu/kubeconfig
```

1.2.7. Cài đặt K3s lên agent

```
export AGENT=192.168.10.134
k3sup join --ip $AGENT --server-ip $SERVER --user adminvm
```

```
K10e681e0e0f6de8696bce1acedec18ccff53438f3f23854d707cb24c39c46a7f84::server:cfc5d74eafca574b0879
87e67f979465
[INFO]  Finding release for channel stable
[INFO]  Using v1.26.5+k3s1 as release
[INFO]  Downloading hash https://github.com/k3s-io/k3s/releases/download/v1.26.5+k3s1/sha256sum-
amd64.txt
[INFO]  Downloading binary https://github.com/k3s-io/k3s/releases/download/v1.26.5+k3s1/k3s
[INFO]  Verifying binary download
[INFO]  Installing k3s to /usr/local/bin/k3s
[INFO]  Skipping installation of SELinux RPM
[INFO]  Creating /usr/local/bin/kubectl symlink to k3s
[INFO]  Creating /usr/local/bin/crictl symlink to k3s
[INFO]  Creating /usr/local/bin/ctr symlink to k3s
[INFO]  Creating killall script /usr/local/bin/k3s-killall.sh
[INFO]  Creating uninstall script /usr/local/bin/k3s-agent-uninstall.sh
[INFO]  env: Creating environment file /etc/systemd/system/k3s-agent.service.env
[INFO]  systemd: Creating service file /etc/systemd/system/k3s-agent.service
[INFO]  systemd: Enabling k3s-agent unit
Created symlink /etc/systemd/system/multi-user.target.wants/k3s-agent.service → /etc/systemd/sys
tem/k3s-agent.service.
[INFO]  systemd: Starting k3s-agent
```

1.2.8. Kiểm tra danh sách nodes trong Cluster

```
kubectl get node -o wide
```

```
root@ubuntu:/home/ubuntu# kubectl get node -o wide
NAME     STATUS   ROLES                  AGE   VERSION        INTERNAL
-IMAGE              KERNEL-VERSION      CONTAINER-RUNTIME
server   Ready    control-plane,master   83s   v1.26.5+k3s1   192.168.
untu 20.04.6 LTS    5.4.0-149-generic   containerd://1.7.1-k3s1
agent    Ready    <none>                 14s   v1.26.5+k3s1   192.168.
untu 20.04.6 LTS    5.4.0-149-generic   containerd://1.7.1-k3s1
```

2. Triển khai Dashboard

2.1. Cài đặt Kubernetes-Dashboard

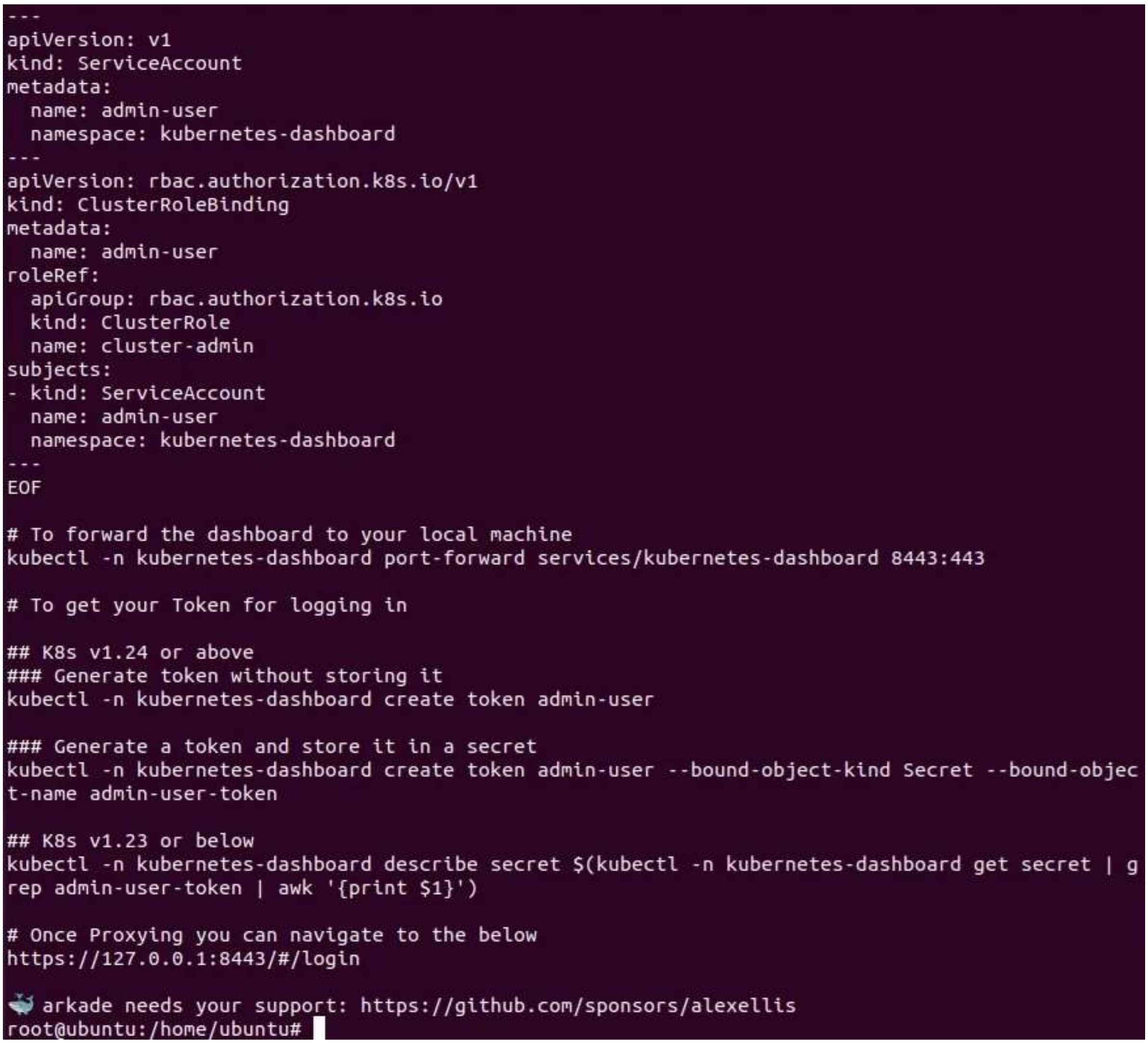
```
arkade install kubernetes-dashboard
```

```
root@ubuntu:/home/ubuntu# arkade install kubernetes-dashboard
Using Kubeconfig: /home/ubuntu/kubeconfig
Node architecture: "amd64"
Using Kubeconfig: /home/ubuntu/kubeconfig
Client: x86_64, Linux
```

2.2. Tạo tài khoản đăng nhập Dashboard

```
arkade install kubernetes-dashboard
cat <<EOF | kubectl apply -f -
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: admin-user
  namespace: kubernetes-dashboard
---
apiVersion: rbac.authorization.k8s.io/v1
```

```
kind: ClusterRoleBinding
metadata:
 name: admin-user
roleRef:
 apiGroup: rbac.authorization.k8s.io
 kind: ClusterRole
 name: cluster-admin
subjects:
- kind: ServiceAccount
 name: admin-user
 namespace: kubernetes-dashboard
---
EOF
```



```
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: admin-user
  namespace: kubernetes-dashboard
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: admin-user
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: cluster-admin
subjects:
- kind: ServiceAccount
  name: admin-user
  namespace: kubernetes-dashboard
---
EOF

# To forward the dashboard to your local machine
kubectl -n kubernetes-dashboard port-forward services/kubernetes-dashboard 8443:443

# To get your Token for logging in

## K8s v1.24 or above
### Generate token without storing it
kubectl -n kubernetes-dashboard create token admin-user

### Generate a token and store it in a secret
kubectl -n kubernetes-dashboard create token admin-user --bound-object-kind Secret --bound-objec
t-name admin-user-token

## K8s v1.23 or below
kubectl -n kubernetes-dashboard describe secret $(kubectl -n kubernetes-dashboard get secret | g
rep admin-user-token | awk '{print $1}')

# Once Proxying you can navigate to the below
https://127.0.0.1:8443/#/login

🐳 arkade needs your support: https://github.com/sponsors/alexellis
root@ubuntu:/home/ubuntu#
```

```
root@ubuntu:/home/ubuntu# cat <<EOF | kubectl apply -f -
---
apiVersion: v1
kind: ServiceAccount
metadata:
  name: admin-user
  namespace: kubernetes-dashboard
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: admin-user
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: cluster-admin
subjects:
- kind: ServiceAccount
  name: admin-user
  namespace: kubernetes-dashboard
---
EOF
serviceaccount/admin-user created
clusterrolebinding.rbac.authorization.k8s.io/admin-user created
root@ubuntu:/home/ubuntu#
```

2.3. Bật Dashboard proxy

- Bật Dashboard proxy để truy cập Dashboard trên máy hiện hành:

```
kubectl proxy
```

2.4. Lấy token đăng nhập Dashboard

```
kubectl -n kubernetes-dashboard create token admin-user
kubectl get pods --all-namespaces
kubectl -n kubernetes-dashboard port-forward kubernetes-dashboard-
9c8547859-wsjb5 8443:8443
```

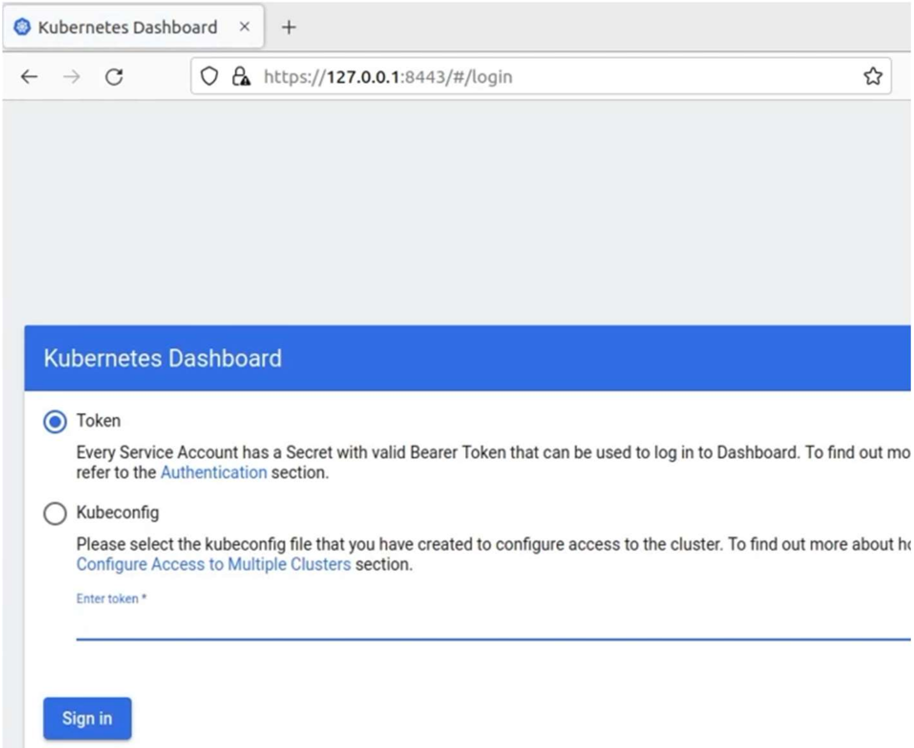
```
ubuntu@ubuntu:~$ sudo su
[sudo] password for ubuntu:
root@ubuntu:/home/ubuntu# kubectl -n kubernetes-dashboard create token admin-user
eyJhbGciOiJSUzI1NiIsImtpZCI6IjFuY3ZLUUdHVTFQcVBVX0pfdTBuS3EwS3ZDQTFxQkZZajZlSlYzdHJCTTAifQ.eyJhd
WQiOlsiaHR0cHM6Ly9rdWJlcm5ldGVzLmRlZmF1bHQuc3ZjLmNsdXN0ZXIubG9jYWwiLCJrM3MiXSwiZXhwIjoxNjg1NjgxN
DY4LCJpYXQiOjE2ODU2Nzc4NjgsImlzcyI6Imh0dHBzOi8va3ViZXJuZXRlcy5kZWZhdWx0LnN2Yy5jbHVzdGVyLmxvY2FsI
iwia3ViZXJuZXRlcy5pbyI6eyJuYW1lc3BhY2UiOiJrdWJlcm5ldGVzLWRhc2hib2FyZCIsInNlcnZpY2VhY2NvdW50Ijp7I
m5hbWUiOiJhZG1pbi11c2VyIiwidWlkIjoiYWIzMmZmZWMtNzhiZS00MTZkLWJlNGUtYTdlOGZhOTJjZjBhIn19LCJuYmYiO
jE2ODU2Nzc4NjgsInN1YiI6InN5c3RlbTpzZXJ2aWNlYWNjb3VudDprdWJlcm5ldGVzLWRhc2hib2FyZDphZG1pbi11c2VyI
n0.nvbLJrVrcQi0zQ5W9WanD-S6l2A4MGjkKhw0AtblThLqez8CuGbySeTR82zB4lUmGXJHhsIc5kxQvLOg8zgAD4-jbqPcy
_Ak1FscXqVWjSmO1MmI_MhNbYBS3jsRjeNWECmd_SEcEfM7rkmxF2RDCaCmIoRqvVt8t01zsbrErMV32eEQDRK8mz8RLSlGF
33qIbWcdQivfaj0JCAOMvVFdnwu9dynyQAHfCXMBfen-KclWzBmETYH-AjMu7EoLvFW69e_j0H-rZ1OaEZrP7EjwQx9vl02J
qg5MK8na0mHemISrV91dJQB9BcQbF2OL92J65W32pOtk61JazjuljPwTQ
```

```
root@ubuntu:/home/ubuntu# kubectl get pods --all-namespaces
NAMESPACE              NAME                                      READY   STATUS      RESTARTS   AGE
kube-system            coredns-59b4f5bbd5-9vzhz                  1/1     Running     0          2m13s
kube-system            local-path-provisioner-76d776f6f9-vzzl9   1/1     Running     0          2m13s
kube-system            helm-install-traefik-crd-rlnk2            0/1     Completed   0          2m13s
kube-system            metrics-server-7b67f64457-x6x48           1/1     Running     0          2m13s
kube-system            helm-install-traefik-k7xzz                0/1     Completed   2          2m13s
kube-system            svclb-traefik-29bd5c0c-jc5cz              2/2     Running     0          76s
kube-system            svclb-traefik-29bd5c0c-b558p              2/2     Running     0          76s
kube-system            traefik-57c84cf78d-tcgrt                  1/1     Running     0          76s
kubernetes-dashboard   kubernetes-dashboard-9c8547859-wsjb5      1/1     Running     0          47s
root@ubuntu:/home/ubuntu#
```

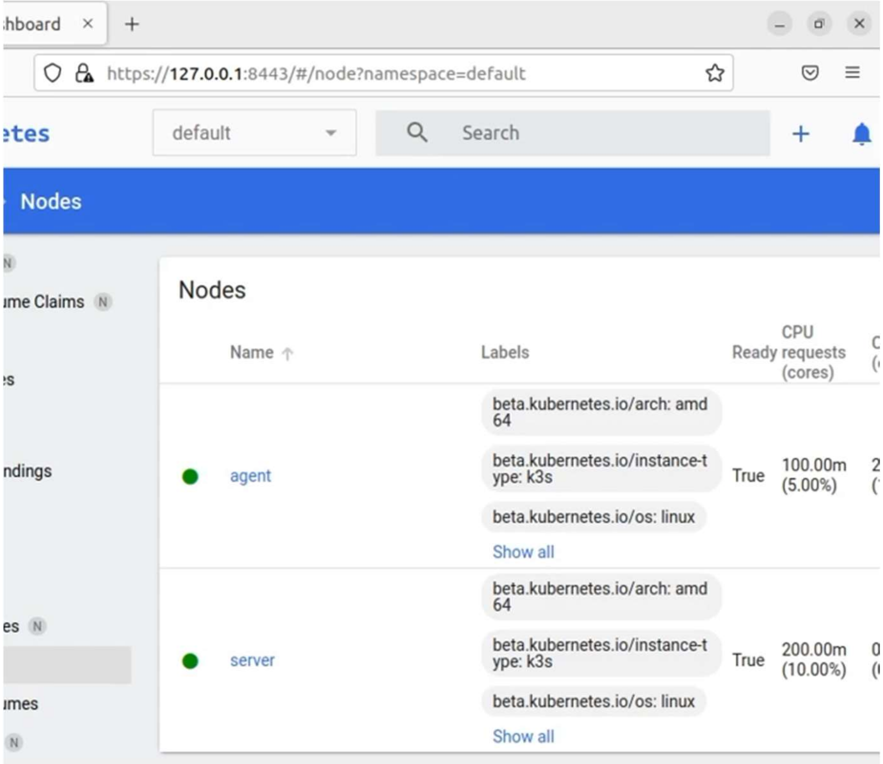
2.5. Truy cập Dashboard từ trình duyệt trên máy hiện hành

- Đăng nhập bằng token:

https://127.0.0.1:8443/



- Kiểm tra trạng thái 2 node vừa tạo:


hboard
https://127.0.0.1:8443/#/node?namespace=default
etes
default
Search
Nodes
ime Claims
ndings
es
imes
Nodes
Name
Labels
Ready
CPU requests (cores)
beta.kubernetes.io/arch: amd64
beta.kubernetes.io/instance-type: k3s
beta.kubernetes.io/os: linux
Show all
agent
True
100.00m (5.00%)
server
True
200.00m (10.00%)

HẾT./.